

Project Documentation

MediScan AI

Multimodal Disease Detection & Guidance System

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Project Name

MediScan AI: Multimodal Disease Detection & Guidance System

Background (Purpose + Problem Statement)

In many rural and semi-urban areas, access to medical professionals is limited, leading to delayed diagnosis and improper treatment. Patients often rely on self-medication or unverified advice, which can worsen health conditions.

The purpose of this project is to develop an AI-powered assistant that can analyze symptoms (text input) and medical images (e.g., skin conditions) to provide preliminary diagnosis and guidance. This system aims to bridge the gap between patients and healthcare services by offering quick, accessible, and reliable assistance.

Objectives (Main Goals)

- Build a multimodal AI system that accepts both image and text inputs
- Classify diseases using machine learning and deep learning
- Provide treatment suggestions and precautions
- Implement explainable AI (XAI)
- Ensure efficiency in low-resource environments

Project Scope

- Focus on skin diseases and common symptoms
- Mobile-friendly web-based interface
- Support low internet or offline usage
- Decision-support system (not a replacement for doctors)

Proposed Solution

The system consists of:

1. Image Processing Module (CNN-based classification)
2. Text-based Symptom Analysis (NLP model)
3. Multimodal Fusion Layer
4. Recommendation System (optional RAG)
5. Explainability Module (Grad-CAM / SHAP)

Comparison with Existing Projects + References

Existing systems are either image-based or text-based. This project combines both modalities.

References:

<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

<https://towardsdatascience.com/skin-cancer-detection>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8382232/>

Datasets

HAM10000 Dataset:

<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

Disease Symptom Dataset:

<https://www.kaggle.com/datasets/itachi9604/disease-symptom-description-dataset>

MedQuAD Dataset:

<https://huggingface.co/datasets/medquad>

Resources

GitHub (to be created): <https://github.com/your-username/MediScan-AI>

Reference Repositories:

<https://github.com/anishathalye/skin-cancer-classification>

<https://github.com/slundberg/shap>

Dashboard: <https://streamlit.io>

Deliverables

- Trained ML/DL models
- Multimodal prediction system
- Web-based interface
- Explainability visualizations
- Final report and GitHub repository
- Demo video and presentation